

IPv6 Protocol I IOT:

IPv4 is the fourth version of the Internet Protocol, and it is the most widely used protocol for network communication. It defines how data is sent and received over the internet.

- ☐ IPv4 addresses are 32-bit numbers represented in **dotted decimal notation** (e.g., 192.168.1.1).
- ☐ Each address consists of four 8-bit octets separated by dots.
- ☐ Common classes:
 - **Class A:** For large networks (1.0.0.0 to 126.0.0.0)
 - **Class B:** For medium-sized networks (128.0.0.0 to 191.255.0.0)
 - **Class C:** For small networks (192.0.0.0 to 223.255.255.0)
- ☐ **Class D** is reserved for multicast, and **Class E** is for experimental purposes.

Private and Public Addresses

- **Private IP ranges:**
 - Class A: 10.0.0.0 – 10.255.255.255
 - Class B: 172.16.0.0 – 172.31.255.255
 - Class C: 192.168.0.0 – 192.168.255.255
- Private addresses are not routable on the public internet and require NAT for communication with external networks.

Transition to IPv6

IPv6 (Internet Protocol version 6) is the most recent version of the Internet Protocol (IP), designed to replace IPv4, which has been in use since the early days of the internet. IPv6 addresses several limitations of IPv4, particularly the exhaustion of available IPv4 addresses, and introduces new features to improve performance, security, and scalability.

Larger Address Space:

- IPv6 uses 128-bit addresses, compared to IPv4's 32-bit addresses.
- This provides approximately 2^{128} unique IP addresses, a significant increase over IPv4's 2^{32} .

Address Representation:

Components in IPv6 Address Format

- There are 8 groups and each group represents 2 Bytes (16-bits).

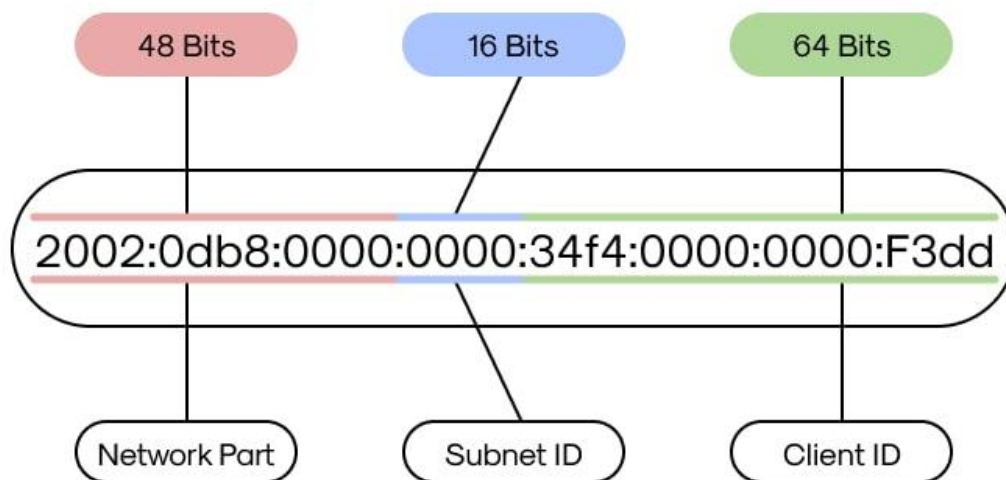
- Each Hex-Digit is of 4 bits (1 nibble)
- Delimiter used – colon (:))



- IPv6 addresses are written as eight groups of four hexadecimal digits, separated by colons (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334).
- Leading zeros in a group can be omitted, and consecutive groups of zeros can be replaced with `::` (e.g., 2001:db8::8a2e:370:7334).



Internet Protocol version 6 address



IPv6 Solution: IPv6 uses 128-bit addresses, supporting approximately 3.4×10^{38} unique addresses. This vast address space ensures scalability for the foreseeable future.

Increased Connectivity:

- The growth of the **Internet of Things (IoT)**, where billions of devices require unique IP addresses, cannot be sustained with IPv4.
- IPv6 enables seamless global addressing, removing the need for techniques like Network Address Translation (NAT) that IPv4 relies on.

Better Support for Modern Applications

- **IPv6 allows for:**
 - More efficient multicast communication.
 - Better Quality of Service (QoS) support, enabling optimized streaming, VoIP, and other real-time applications.
 - Simplified peer-to-peer communications without the need for NAT traversal.
 - **Better Header Format:** IPv6 uses a new header format in which options are separated from the base header and inserted, when needed, between the base header and the upper layer data. This simplifies and speeds up the routing process
 - **Support For More Security:** The encryption and authentication options in IPv6 provide confidentiality and integrity of the packet. In IPv6 representation, we have three addressing methods :
 - Unicast
 - Multicast
 - Anycast
- Addressing Methods

Unicast Address :

- Unicast Address identifies a single network interface. A packet sent to a unicast address is delivered to the interface identified by that address.

Multicast Address :

Multicast Address is used by multiple hosts, called as groups, acquires a multicast destination address. These hosts need not be geographically together. If any packet is sent to this multicast address, it will be distributed to all interfaces corresponding to that multicast address. And every node is configured in the same way. In simple words, one data packet is sent to multiple destinations simultaneously.

Anycast Address:

- Anycast Address is assigned to a group of interfaces. Any packet sent to an anycast address will be delivered to only one member interface (mostly nearest host possible).

Advantages of IPv6

- **1. Realtime Data Transmission :** Realtime data transmission refers to the process of transmitting data in a very fast manner or **immediately**. Example : Live streaming

services such as cricket matches, or other tournament that are streamed on web exactly as soon as it happens with a maximum delay of 5-6 seconds.

- **2. IPv6 supports authentication:** Verifying that the data received by the receiver from the sender is exactly what the sender sent and came through the sender only not from any third party. Example : Matching the hash value of both the messages for verification is also done by IPv6.
- **3. IPv6 performs Encryption:** Ipv6 can encrypt the message at network layer even if the protocols of application layer at user level didn't encrypt the message which is a major advantage as it takes care of encryption.
- **4. Faster processing at Router:** Routers are able to process data packets of Ipv6 much faster due to smaller **Base header** of fixed size – 40 bytes which helps in decreasing processing time resulting in more efficient packet transmission. Whereas in Ipv4, we have to calculate the length of header which lies between 20-60 bytes